

Response to your email – point by point

Each concern + our response + a real figure that backs it.

POINT 1

Profile HMMs capture the alignment of multiple sequences ... so this might be useful for checking if a new copy belongs to a family, but I don't see how this can help you in assigning a read to a paralog.

OUR RESPONSE

You are correct. A classical profile HMM built from an MSA of all copies absorbs SNP variation into degenerate match states (e.g., 0.50 / 0.50 at every SNP column). One model → no DlogL across copies → cannot assign.

We never meant a classical profile HMM. The figure on the right shows the contrast: a family HMM is degenerate; per-copy HMMs are not.

Evidence: fig_hmm4_family_vs_copy.png

POINT 2

I also don't understand your idea of building a profile HMM per copy, and how you would use it to solve the assignment problem.

OUR RESPONSE

A 'per-copy HMM' is a per-position emission model derived from one copy's transcript sequence:

$$P(\text{base} = S_c[i] \mid \text{copy } c) = 0.95$$

$$P(\text{other base} \mid \text{copy } c) = 0.017$$

Evidence: fig_hmm2_scoring.png

For a read R with M matches against copy c (out of L bases):

$$\log L(R \mid \text{copy } c) = M \cdot \log(0.95) + (L - M) \cdot \log(0.017)$$

POINT 3

(Implicit from your email: does this approach actually solve the problem?)

OUR RESPONSE

On real GOLGA81 FLNC IsoSeq reads: 4 / 4 reads → MAPQ = 60 with minimap2 alone. HMM scoring AGREES with minimap2 at the class boundary, never rescues a tie minimap2 cannot resolve. The HMM step is therefore REDUNDANT for our data.

Where minimap2 ties (MAPQ=0), the HMM also ties: Theorem 1 explains why (within-class members are read-indistinguishable).

Evidence: fig_flnc_assignment.png | fig_class_boundary_resolution.png | fig_em_bootstrap.png

POINT 4

Explain these 'in isolation', without mixing them up with other concepts such as variation graphs etc.

OUR RESPONSE

The HMM explanation above (Points 1–2) contains no VG concepts.

The VG is a SEPARATE framework that answers a DIFFERENT question:

'Which copies can in principle be distinguished, and which collapse into one equivalence class by sequence identity?' (resolvability)

Evidence: fig_advisor_floor.png | fig_advisor_pipeline.png

VG does not score reads. HMM does not define resolvability. We presented

BOTTOM LINE

You were right to push back. The HMM step as originally framed adds nothing for FLNC reads on coherent multi-copy families with distinct VG paths.

What does load-bearing work: (i) VG → equivalence classes (the resolvability floor), (ii) minimap2 (alignment + class-aware MAPQ),

(iii) FLNC-length consistency term (closes residual ties from paralog repeats), (iv) EM bootstrap (distributes any remaining ambiguous reads).